organoid

sc

organoid_agg

sc_agg

−log10 FDR

treatment coefficient

R2

1.0

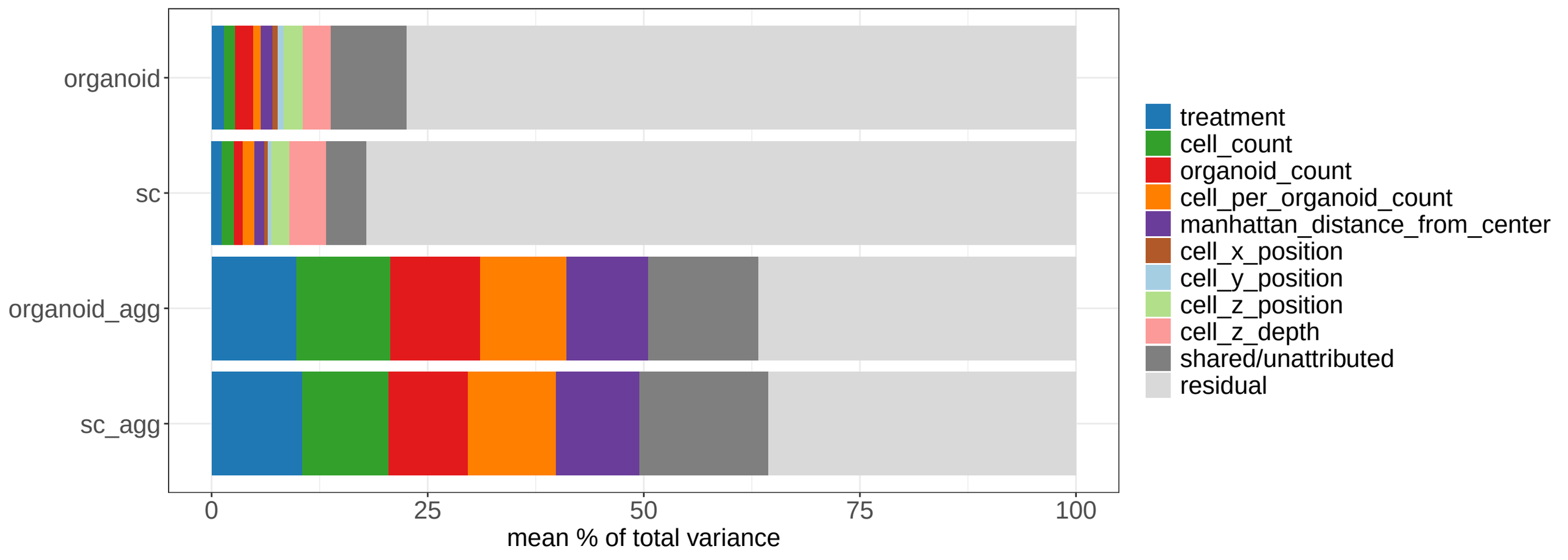
0.8

0.6

0.4

0.2

0.0


organoid
sc
organoid_agg
sc_agg
0
25
50
75
100
mean % of total variance
treatment
cell_count
organoid_count
cell_per_organoid_count
manhattan_distance_from_center
cell_x_position
cell_y_position
cell_z_position
cell_z_depth
shared/unattributed
residual

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

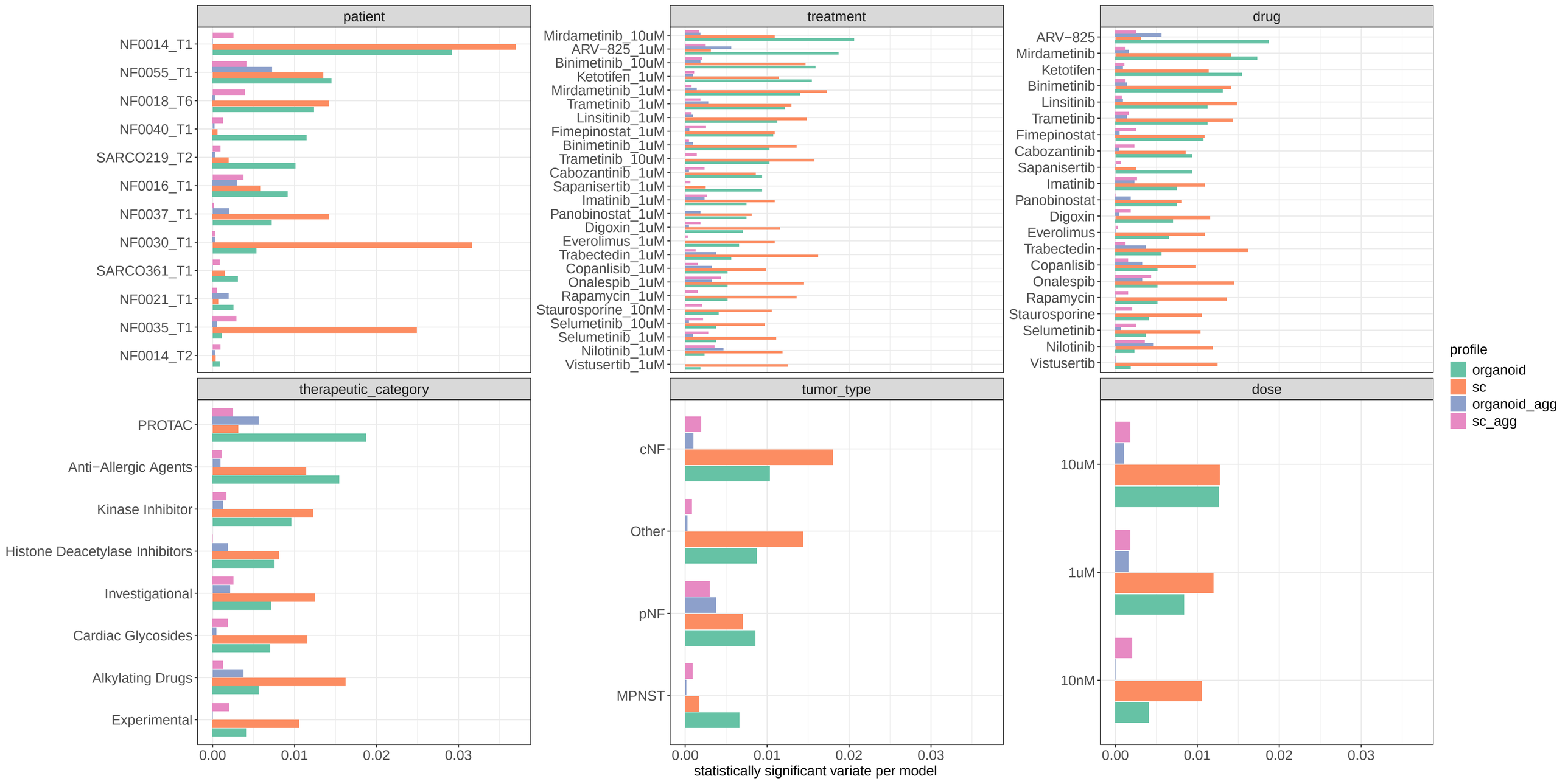
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

**median treatment % var**
3
2
1
0

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

median treatment % var
3
2
1
0


patient
NF0014_T1
NF0055_T1
NF0018_T6
NF0040_T1
SARCO219_T2
NF0016_T1
NF0037_T1
NF0030_T1
SARCO361_T1
NF0021_T1
NF0035_T1
NF0014_T2
treatment
Mirdametinib_10uM
ARV−825_1uM
Binimetinib_10uM
Ketotifen_1uM
Mirdametinib_1uM
Trametinib_1uM
Linsitinib_1uM
Fimepinostat_1uM
Binimetinib_1uM
Trametinib_10uM
Cabozantinib_1uM
Sapanisertib_1uM
Imatinib_1uM
Panobinostat_1uM
Digoxin_1uM
Everolimus_1uM
Trabectedin_1uM
Copanlisib_1uM
Onalespib_1uM
Rapamycin_1uM
Staurosporine_10nM
Selumetinib_10uM
Selumetinib_1uM
Nilotinib_1uM
Vistusertib_1uM
drug
ARV−825
Mirdametinib
Ketotifen
Binimetinib
Linsitinib
Trametinib
Fimepinostat
Cabozantinib
Sapanisertib
Imatinib
Panobinostat
Digoxin
Everolimus
Trabectedin
Copanlisib
Onalespib
Rapamycin
Staurosporine
Selumetinib
Nilotinib
Vistusertib
profile
organoid
sc
organoid_agg
sc_agg
therapeutic_category
PROTAC
Anti−Allergic Agents
Kinase Inhibitor
Histone Deacetylase Inhibitors
Investigational
Cardiac Glycosides
Alkylating Drugs
Experimental
tumor_type
cNF
Other
pNF
MPNST
dose
10uM
1uM
10nM
0.00
0.01
0.02
0.03
statistically significant variate per model

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

**statistically significant variate per model**

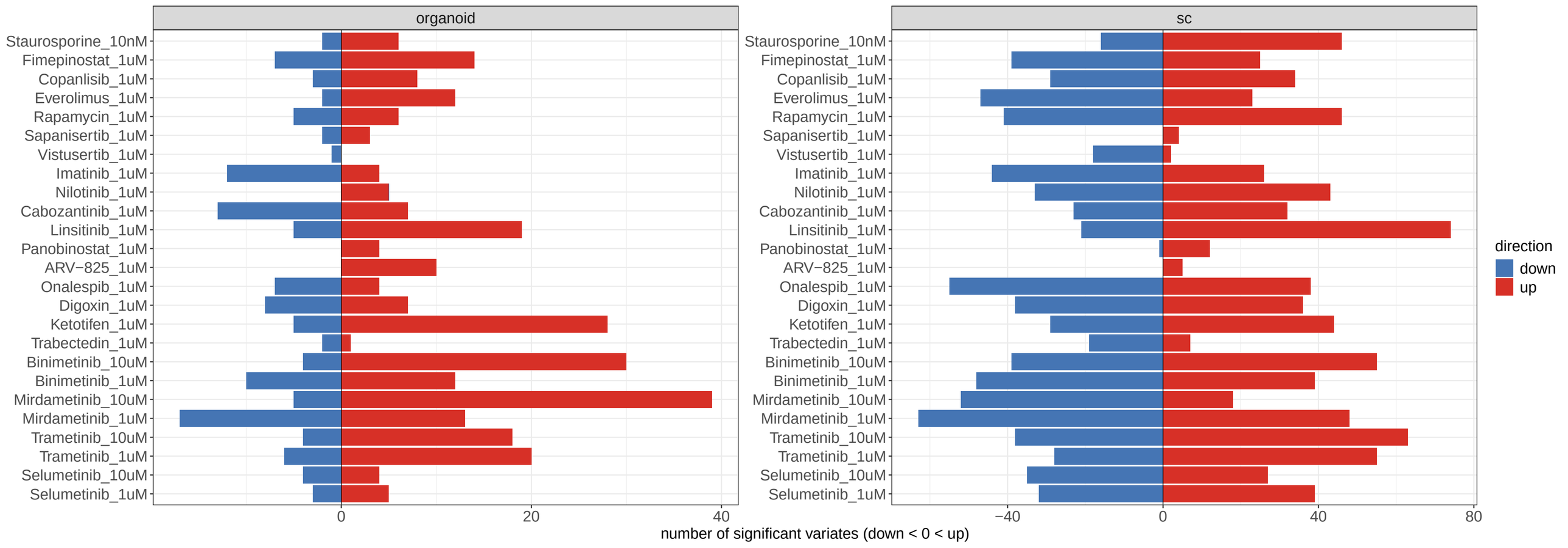
0.15
0.1
0.05
0

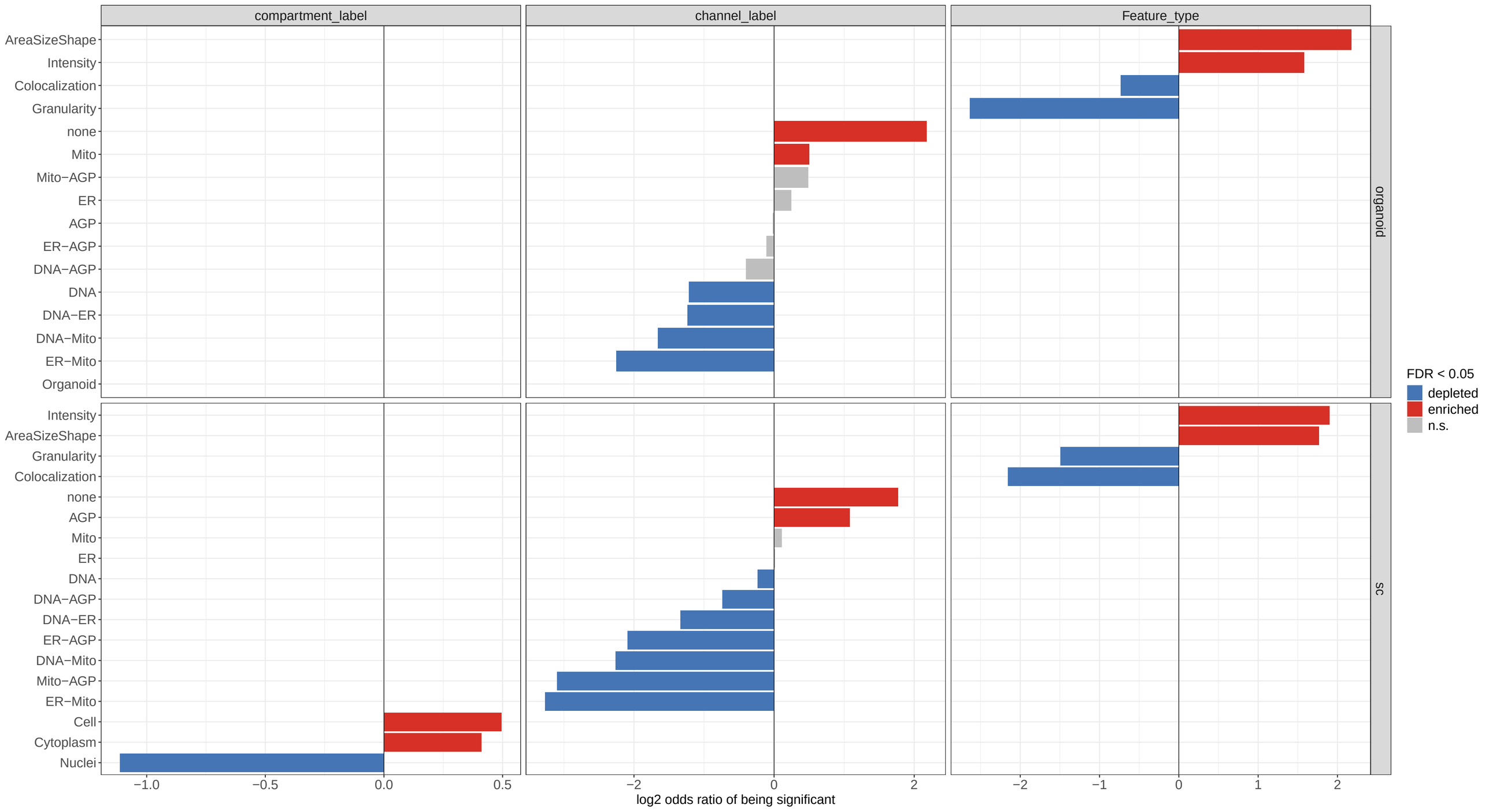
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

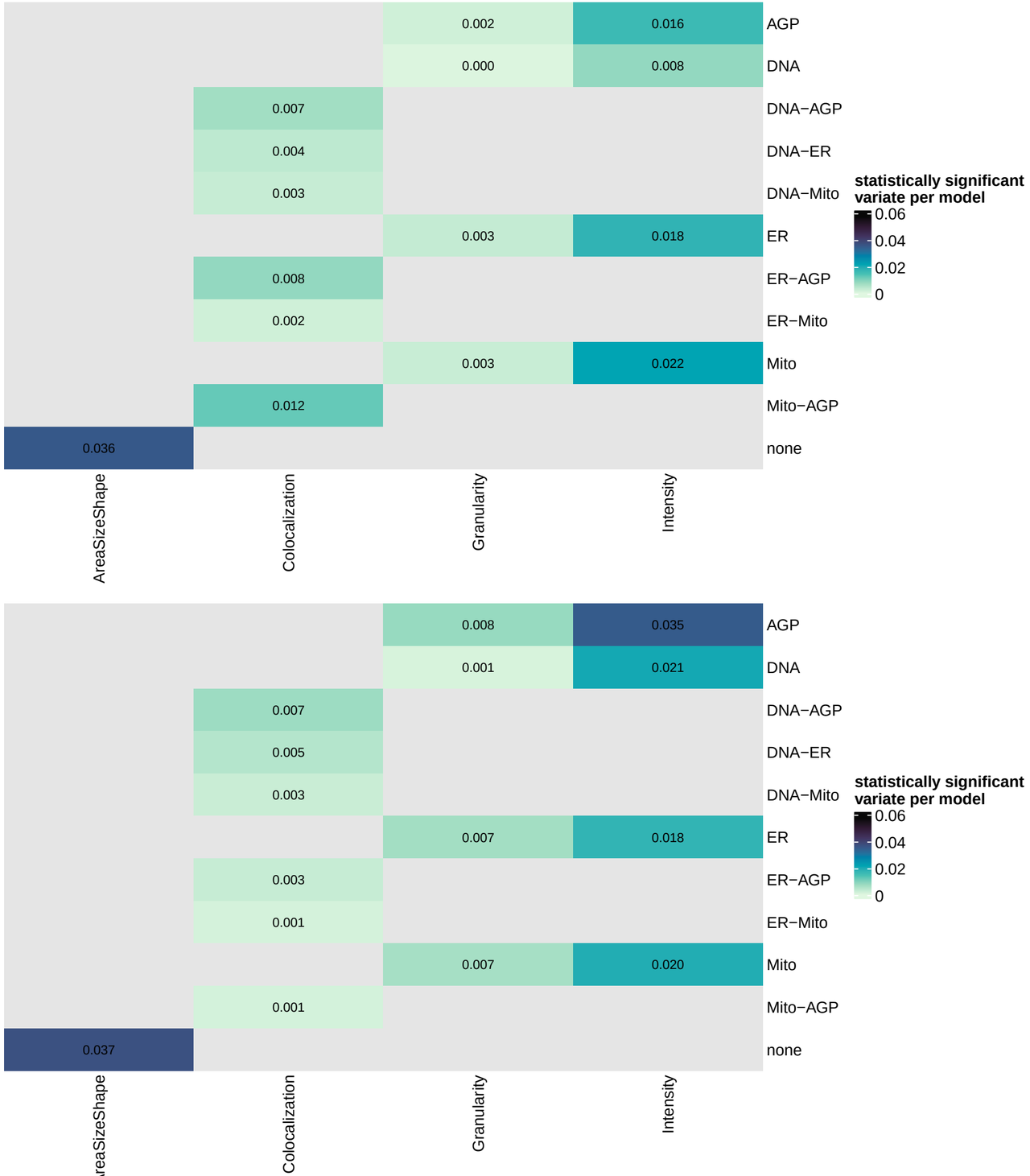
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

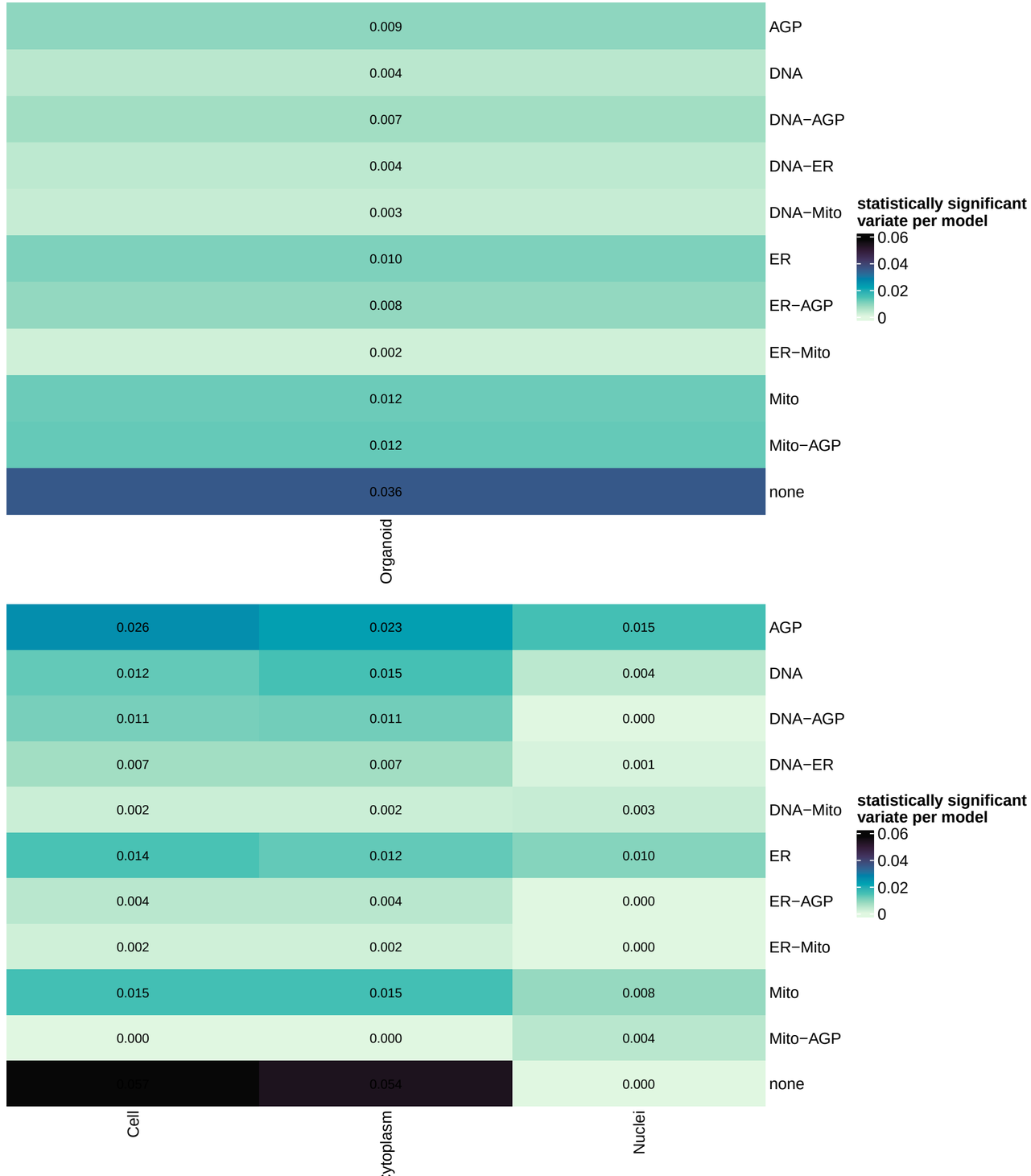
**statistically significant variate per model**

0.15
0.1
0.05
0


organoid
sc
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM
0
20
40
−40
0
40
80
number of significant variates (down < 0 < up)
direction
down
up

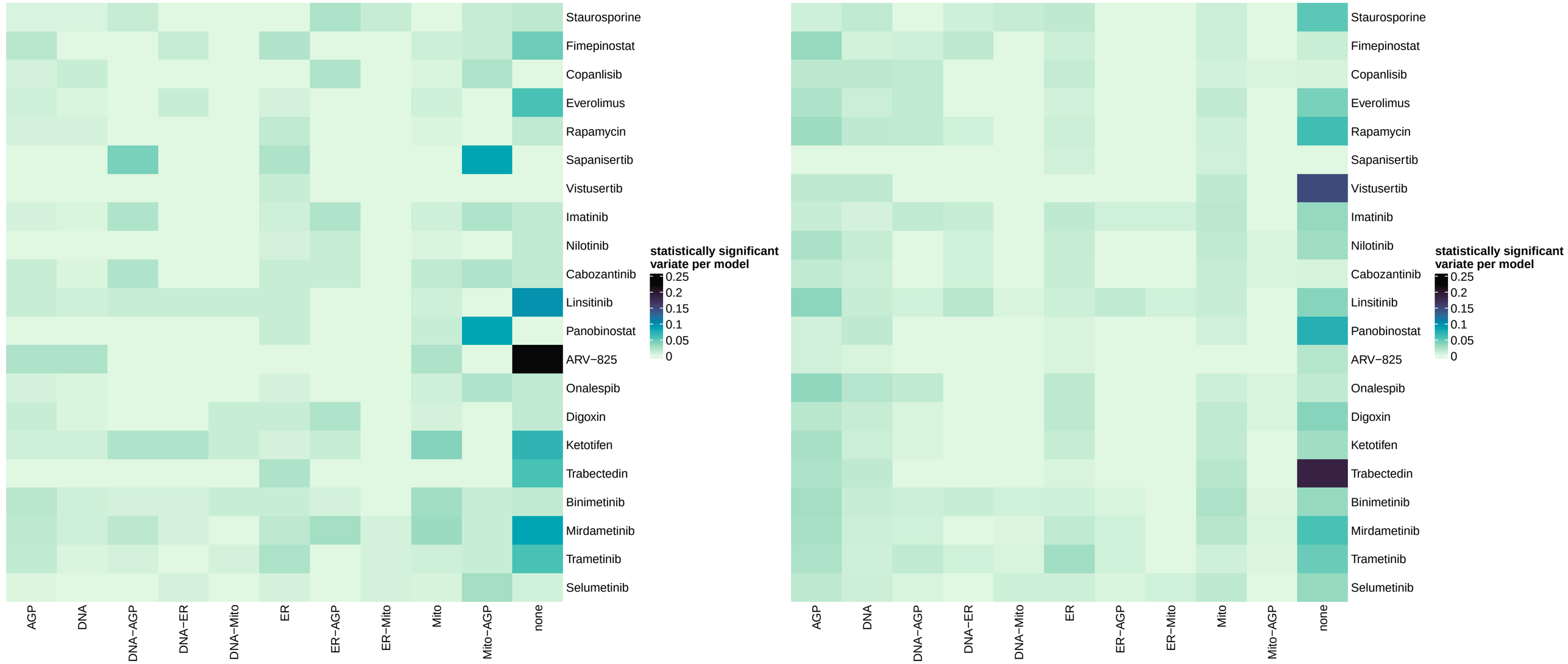

compartment_label
channel_label
Feature_type
organoid
sc
AreaSizeShape
Intensity
Colocalization
Granularity
none
Mito
Mito−AGP
ER
AGP
ER−AGP
DNA−AGP
DNA
DNA−ER
DNA−Mito
ER−Mito
Organoid
Cell
Cytoplasm
Nuclei
−1.0
−0.5
0.0
0.5
−2
0
2
−2
−1
0
1
2
log2 odds ratio of being significant
FDR < 0.05
depleted
enriched
n.s.


0.002
0.016
AGP
0.000
0.008
DNA
0.007
DNA−AGP
0.004
DNA−ER
0.003
DNA−Mito
0.003
0.018
ER
0.008
ER−AGP
0.002
ER−Mito
0.003
0.022
Mito
0.012
Mito−AGP
0.036
none
statistically significant variate per model
0.06
0.04
0.02
0
AreaSizeShape
Colocalization
Granularity
Intensity


0.009
AGP
0.004
DNA
0.007
DNA−AGP
0.004
DNA−ER
0.003
DNA−Mito
0.010
ER
0.008
ER−AGP
0.002
ER−Mito
0.012
Mito
0.012
Mito−AGP
0.036
none
statistically significant variate per model
0.06
0.04
0.02
0
Organoid


0.008
0.035
AGP
0.001
0.021
DNA
0.007
DNA−AGP
0.005
DNA−ER
0.003
DNA−Mito
0.007
0.018
ER
0.003
ER−AGP
0.001
ER−Mito
0.007
0.020
Mito
0.001
Mito−AGP
0.037
none
statistically significant variate per model
0.06
0.04
0.02
0
AreaSizeShape
Colocalization
Granularity
Intensity


0.026
0.023
0.015
AGP
0.012
0.015
0.004
DNA
0.011
0.011
0.000
DNA−AGP
0.007
0.007
0.001
DNA−ER
0.002
0.002
0.003
DNA−Mito
0.014
0.012
0.010
ER
0.004
0.004
0.000
ER−AGP
0.002
0.002
0.000
ER−Mito
0.015
0.015
0.008
Mito
0.000
0.000
0.004
Mito−AGP
0.057
0.054
0.000
none
statistically significant variate per model
0.06
0.04
0.02
0
Cell
Cytoplasm
Nuclei


Staurosporine
Fimepinostat
Copanlisib
Everolimus
Rapamycin
Sapanisertib
Vistusertib
Imatinib
Nilotinib
Cabozantinib
Linsitinib
Panobinostat
ARV−825
Onalespib
Digoxin
Ketotifen
Trabectedin
Binimetinib
Mirdametinib
Trametinib
Selumetinib
AGP
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
Mito−AGP
none
statistically significant variate per model
0.25
0.2
0.15
0.1
0.05
0

| | Binimetinib_10uM | Binimetinib_1uM | Mirdametinib_10uM | Mirdametinib_1uM | Trametinib_10uM | Trametinib_1uM | Selumetinib_10uM | Selumetinib_1uM |
|---|---|---|---|---|---|---|---|---|
| Binimetinib_10uM | 1.00 | 0.19 | 0.24 | 0.21 | 0.27 | 0.21 | 0.28 | 0.10 |
| Binimetinib_1uM | 0.19 | 1.00 | 0.21 | 0.25 | 0.19 | 0.20 | 0.21 | 0.21 |
| Mirdametinib_10uM | 0.24 | 0.21 | 1.00 | 0.14 | 0.19 | 0.17 | 0.23 | 0.17 |
| Mirdametinib_1uM | 0.21 | 0.25 | 0.14 | 1.00 | 0.21 | 0.24 | 0.25 | 0.25 |
| Trametinib_10uM | 0.27 | 0.19 | 0.19 | 0.21 | 1.00 | 0.33 | 0.35 | 0.37 |
| Trametinib_1uM | 0.21 | 0.20 | 0.17 | 0.24 | 0.33 | 1.00 | 0.23 | 0.27 |
| Selumetinib_10uM | 0.28 | 0.21 | 0.23 | 0.25 | 0.35 | 0.23 | 1.00 | 0.34 |
| Selumetinib_1uM | 0.10 | 0.21 | 0.17 | 0.25 | 0.37 | 0.27 | 0.34 | 1.00 |

Spearman: 1, 0.5, 0, −0.5, −1

| | Binimetinib_10uM | Binimetinib_1uM | Mirdametinib_10uM | Mirdametinib_1uM | Trametinib_10uM | Trametinib_1uM | Selumetinib_10uM | Selumetinib_1uM |
|---|---|---|---|---|---|---|---|---|
| Binimetinib_10uM | 1.00 | 0.25 | 0.18 | 0.34 | 0.25 | 0.25 | 0.24 | 0.13 |
| Binimetinib_1uM | 0.25 | 1.00 | 0.17 | 0.14 | 0.15 | 0.23 | 0.11 | 0.11 |
| Mirdametinib_10uM | 0.18 | 0.17 | 1.00 | 0.18 | 0.20 | 0.16 | 0.32 | 0.15 |
| Mirdametinib_1uM | 0.34 | 0.14 | 0.18 | 1.00 | 0.21 | 0.25 | 0.35 | 0.25 |
| Trametinib_10uM | 0.25 | 0.15 | 0.20 | 0.21 | 1.00 | 0.30 | 0.28 | 0.29 |
| Trametinib_1uM | 0.25 | 0.23 | 0.16 | 0.25 | 0.30 | 1.00 | 0.23 | 0.20 |
| Selumetinib_10uM | 0.24 | 0.11 | 0.32 | 0.35 | 0.28 | 0.23 | 1.00 | 0.31 |
| Selumetinib_1uM | 0.13 | 0.11 | 0.15 | 0.25 | 0.29 | 0.20 | 0.31 | 1.00 |

Spearman: 1, 0.5, 0, −0.5, −1

| | Binimetinib_10uM | Binimetinib_1uM | Mirdametinib_10uM | Mirdametinib_1uM | Trametinib_10uM | Trametinib_1uM | Selumetinib_10uM | Selumetinib_1uM |
|---|---|---|---|---|---|---|---|---|
| Binimetinib_10uM | 1.00 | 0.34 | 0.32 | 0.32 | 0.35 | 0.23 | 0.27 | 0.14 |
| Binimetinib_1uM | 0.34 | 1.00 | 0.21 | 0.30 | 0.23 | 0.23 | 0.22 | 0.18 |
| Mirdametinib_10uM | 0.32 | 0.21 | 1.00 | 0.13 | 0.21 | 0.13 | 0.26 | 0.15 |
| Mirdametinib_1uM | 0.32 | 0.30 | 0.13 | 1.00 | 0.29 | 0.34 | 0.26 | 0.25 |
| Trametinib_10uM | 0.35 | 0.23 | 0.21 | 0.29 | 1.00 | 0.27 | 0.20 | 0.27 |
| Trametinib_1uM | 0.23 | 0.23 | 0.13 | 0.34 | 0.27 | 1.00 | 0.19 | 0.31 |
| Selumetinib_10uM | 0.27 | 0.22 | 0.26 | 0.26 | 0.20 | 0.19 | 1.00 | 0.23 |
| Selumetinib_1uM | 0.14 | 0.18 | 0.15 | 0.25 | 0.27 | 0.31 | 0.23 | 1.00 |

Spearman: 1, 0.5, 0, −0.5, −1

| | Binimetinib_10uM | Binimetinib_1uM | Mirdametinib_10uM | Mirdametinib_1uM | Trametinib_10uM | Trametinib_1uM | Selumetinib_10uM | Selumetinib_1uM |
|---|---|---|---|---|---|---|---|---|
| Binimetinib_10uM | 1.00 | 0.30 | 0.23 | 0.31 | 0.27 | 0.22 | 0.25 | 0.09 |
| Binimetinib_1uM | 0.30 | 1.00 | 0.21 | 0.21 | 0.22 | 0.26 | 0.19 | 0.13 |
| Mirdametinib_10uM | 0.23 | 0.21 | 1.00 | 0.30 | 0.29 | 0.20 | 0.37 | 0.26 |
| Mirdametinib_1uM | 0.31 | 0.21 | 0.30 | 1.00 | 0.24 | 0.23 | 0.40 | 0.20 |
| Trametinib_10uM | 0.27 | 0.22 | 0.29 | 0.24 | 1.00 | 0.29 | 0.35 | 0.26 |
| Trametinib_1uM | 0.22 | 0.26 | 0.20 | 0.23 | 0.29 | 1.00 | 0.28 | 0.17 |
| Selumetinib_10uM | 0.25 | 0.19 | 0.37 | 0.40 | 0.35 | 0.28 | 1.00 | 0.33 |
| Selumetinib_1uM | 0.09 | 0.13 | 0.26 | 0.20 | 0.26 | 0.17 | 0.33 | 1.00 |

Spearman: 1, 0.5, 0, −0.5, −1